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### Transitioning, With A Peripheral Device, Between User Devices Within A Video Conference

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#### Abstract

A method including transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference. Performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference. Disconnecting the first user device from the video conference before the second user device is connected to the video conference. Transitioning the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. application Ser. No. 18/193,319, filed Mar. 30, 2023, the entire disclosure of which is herein incorporated by reference.

### FIELD

[0002] This disclosure relates generally to video conferencing and, more specifically, to using a peripheral device to transition between user devices within a video conference.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0003] This disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

[0004] FIG. 1 is a block diagram of an example of an electronic computing and communications system.

[0005] FIG. 2 is a block diagram of an example internal configuration of a computing device of an electronic computing and communications system.

[0006] FIG. 3 is a block diagram of an example of a software platform implemented by an electronic computing and communications system.

[0007] FIG. 4 is a block diagram of an example of a system for using a peripheral device to transition between user devices within a video conference.

[0008] FIG. 5 is a block diagram of an example of transitioning between user devices.

[0009] FIG. 6 is a block diagram of an example of timing associated with transitioning between user devices.

[0010] FIG. 7 is an illustration of an example of a graphical user interface (GUI) output for display during a video conference.

[0011] FIG. 8 is a flowchart of an example of a technique for buffering a portion of media content in a peripheral device.

[0012] FIG. 9 is a flowchart of an example of a technique for using a peripheral device to transition between user devices within a video conference.

### DETAILED DESCRIPTION OF THE DRAWINGS

[0013] Enterprise entities rely upon several modes of communication to support their operations, including telephone, email, internal messaging, and the like. These separate modes of communication have historically been implemented by service providers whose services are not integrated with one another. The disconnect between these services, in at least some cases, requires information to be manually passed by users from one service to the next. Furthermore, some services, such as telephony services, are traditionally delivered via on-premises systems, meaning that remote workers and those who are generally increasingly mobile may be unable to rely upon them. One type of system which addresses problems such as these includes a unified communications as a service (UCaaS) platform, which includes several communications services integrated over a network, such as the Internet, to deliver a complete communication experience

regardless of physical location.

[0014] Individuals may use software such as of a UCaaS platform to communicate and collaborate remotely with one another in video conferences. During a video conference, a conference participant may use a device (e.g., a desktop, laptop, tablet, or other computer, including a display, camera, microphone, and speakers) to send video data depicting themselves and audio data representing their speech to other conference participants and to receive video and audio data from the other conference participants. In some cases, during a video conference, a conference participant may wish to transition from one device to another to continue participating in the video conference. For example, the conference participant could be using a smartphone while traveling to a location (e.g., while walking or a passenger in a vehicle), then may wish to transition to a desktop, laptop, or tablet at the location. In another example, the conference participant could be using a device associated with a particular space, such as a computer in an office or living room, then may wish to transition to another device associated with another space, such as another computer in a conference room or kitchen. During the transition from one device to another, the conference participant may temporarily disconnect from the video conference. As a result, the conference participant may lose information associated with the video conference due to the disconnection. For example, the disconnection may cause the conference participant to miss media content and lose chat messages shared during the video conference.

[0015] Implementations of this disclosure address problems such as these by designating a peripheral device to buffer a portion of media content of a video conference and by causing the peripheral device to adaptively relay the portion of the media content to a conference participant while transitioning between devices connected to the video conference. A first user device (e.g., a desktop, laptop, tablet, smartphone or other computer) associated with a conference participant may connect to a video conference and may transmit and receive media content during the video conference (e.g., sending and receiving video and/or audio data). The first user device may also connect to a peripheral device for relaying the media content to the conference participant. For example, the peripheral device could be a device configured to present video and/or audio data to the conference participant, such as a wearable device (e.g., headphones, headset, smartwatch, or glasses) or another device (e.g., display or speakers).

[0016] The first user device may transmit, to the peripheral device, an indication of an upcoming transition or switch at some point during the video conference. The indication may cause the peripheral device to temporarily buffer a portion of the media content (e.g., storing less than a predefined amount time of the media content in a micro-buffer, such as less than five seconds of the media content in a random access memory (RAM) of the peripheral device) (e.g., the peripheral device may temporarily buffer a portion of the video conference in preparation for the first user device to disconnect from the video conference). In some implementations, the portion of the media content may include content other than video and/or audio media shared by participants to the video conference, such as chat messages or metadata associated with the video conference.

[0017] The system may then perform, within the video conference, a transition of the conference participant from the first user device connected to the video conference to a second user device connected to the video conference (e.g., another desktop, laptop, tablet, smartphone, or other computer), such as by communicating a handoff event between the first user device and the second user device. Performing the transition may include disconnecting the first user device from the video conference, connecting the second user device to the video conference), and terminating the communication between the first user device and the second user device. During the transition, the peripheral device may relay the portion of the media content from the buffer to the conference participant. The peripheral device may adaptively relay the portion of the media content to the conference participant, such as by reducing video quality and/or audio quality of the portion of the media content in a controlled manner. The peripheral device may perform a time alignment between the portion of the media content and the media content received by the second user device

and then resume a real-time relay of media content using the second user device. As a result, the peripheral device can relay the media content to a conference participant during a transition between devices while reducing loss of information.

[0018] To describe some implementations in greater detail, reference is first made to examples of hardware and software structures used to implement a system for using a peripheral device to transition between user devices within a video conference. FIG. 1 is a block diagram of an example of an electronic computing and communications system **100**, which can be or include a distributed computing system (e.g., a client-server computing system), a cloud computing system, a clustered computing system, or the like.

[0019] The system **100** includes one or more customers, such as customers **102A** through **102B**, which may each be a public entity, private entity, or another corporate entity or individual that purchases or otherwise uses software services, such as of a UCaaS platform provider. Each customer can include one or more clients. For example, as shown and without limitation, the customer **102A** can include clients **104A** through **104B**, and the customer **102B** can include clients **104C** through **104D**. A customer can include a customer network or domain. For example, and without limitation, the clients **104A** through **104B** can be associated or communicate with a customer network or domain for the customer **102A** and the clients **104C** through **104D** can be associated or communicate with a customer network or domain for the customer **102B**.

[0020] A client, such as one of the clients **104A** through **104D**, may be or otherwise refer to one or both of a client device or a client application. Where a client is or refers to a client device, the client can comprise a computing system, which can include one or more computing devices, such as a mobile phone, a tablet computer, a laptop computer, a notebook computer, a desktop computer, or another suitable computing device or combination of computing devices. Where a client instead is or refers to a client application, the client can be an instance of software running on a customer device (e.g., a client device or another device). In some implementations, a client can be implemented as a single physical unit or as a combination of physical units. In some implementations, a single physical unit can include multiple clients.

[0021] The system **100** can include a number of customers and/or clients or can have a configuration of customers or clients different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include hundreds or thousands of customers, and at least some of the customers can include or be associated with a number of clients.

[0022] The system **100** includes a datacenter **106**, which may include one or more servers. The datacenter **106** can represent a geographic location, which can include a facility, where the one or more servers are located. The system **100** can include a number of datacenters and servers or can include a configuration of datacenters and servers different from that generally illustrated in FIG. 1. For example, and without limitation, the system **100** can include tens of datacenters, and at least some of the datacenters can include hundreds or another suitable number of servers. In some implementations, the datacenter **106** can be associated or communicate with one or more datacenter networks or domains, which can include domains other than the customer domains for the customers **102A** through **102B**.

[0023] The datacenter **106** includes servers used for implementing software services of a UCaaS platform. The datacenter **106** as generally illustrated includes an application server **108**, a database server **110**, and a telephony server **112**. The servers **108** through **112** can each be a computing system, which can include one or more computing devices, such as a desktop computer, a server computer, or another computer capable of operating as a server, or a combination thereof. A suitable number of each of the servers **108** through **112** can be implemented at the datacenter **106**. The UCaaS platform uses a multi-tenant architecture in which installations or instantiations of the servers **108** through **112** is shared amongst the customers **102A** through **102B**.

[0024] In some implementations, one or more of the servers **108** through **112** can be a non-hardware server implemented on a physical device, such as a hardware server. In some

implementations, a combination of two or more of the application server **108**, the database server **110**, and the telephony server **112** can be implemented as a single hardware server or as a single non-hardware server implemented on a single hardware server. In some implementations, the datacenter **106** can include servers other than or in addition to the servers **108** through **112**, for example, a media server, a proxy server, or a web server.

[0025] The application server **108** runs web-based software services deliverable to a client, such as one of the clients **104A** through **104D**. As described above, the software services may be of a UCaaS platform. For example, the application server **108** can implement all or a portion of a UCaaS platform, including conferencing software, messaging software, and/or other intra-party or inter-party communications software. The application server **108** may, for example, be or include a unitary Java Virtual Machine (JVM).

[0026] In some implementations, the application server **108** can include an application node, which can be a process executed on the application server **108**. For example, and without limitation, the application node can be executed in order to deliver software services to a client, such as one of the clients **104A** through **104D**, as part of a software application. The application node can be implemented using processing threads, virtual machine instantiations, or other computing features of the application server **108**. In some such implementations, the application server **108** can include a suitable number of application nodes, depending upon a system load or other characteristics associated with the application server **108**. For example, and without limitation, the application server **108** can include two or more nodes forming a node cluster. In some such implementations, the application nodes implemented on a single application server **108** can run on different hardware servers.

[0027] The database server **110** stores, manages, or otherwise provides data for delivering software services of the application server **108** to a client, such as one of the clients **104A** through **104D**. In particular, the database server **110** may implement one or more databases, tables, or other information sources suitable for use with a software application implemented using the application server **108**. The database server **110** may include a data storage unit accessible by software executed on the application server **108**. A database implemented by the database server **110** may be a relational database management system (RDBMS), an object database, an XML database, a configuration management database (CMDB), a management information base (MIB), one or more flat files, other suitable non-transient storage mechanisms, or a combination thereof. The system **100** can include one or more database servers, in which each database server can include one, two, three, or another suitable number of databases configured as or comprising a suitable database type or combination thereof.

[0028] In some implementations, one or more databases, tables, other suitable information sources, or portions or combinations thereof may be stored, managed, or otherwise provided by one or more of the elements of the system **100** other than the database server **110**, for example, the client **104** or the application server **108**.

[0029] The telephony server **112** enables network-based telephony and web communications from and to clients of a customer, such as the clients **104A** through **104B** for the customer **102A** or the clients **104C** through **104D** for the customer **102B**. Some or all of the clients **104A** through **104D** may be voice over internet protocol (VOIP)-enabled devices configured to send and receive calls over a network **114**. In particular, the telephony server **112** includes a session initiation protocol (SIP) zone and a web zone. The SIP zone enables a client of a customer, such as the customer **102A** or **102B**, to send and receive calls over the network **114** using SIP requests and responses. The web zone integrates telephony data with the application server **108** to enable telephony-based traffic access to software services run by the application server **108**. Given the combined functionality of the SIP zone and the web zone, the telephony server **112** may be or include a cloud-based private branch exchange (PBX) system.

[0030] The SIP zone receives telephony traffic from a client of a customer and directs same to a

destination device. The SIP zone may include one or more call switches for routing the telephony traffic. For example, to route a VOIP call from a first VOIP-enabled client of a customer to a second VOIP-enabled client of the same customer, the telephony server **112** may initiate a SIP transaction between a first client and the second client using a PBX for the customer. However, in another example, to route a VOIP call from a VOIP-enabled client of a customer to a client or non-client device (e.g., a desktop phone which is not configured for VOIP communication) which is not VOIP-enabled, the telephony server **112** may initiate a SIP transaction via a VOIP gateway that transmits the SIP signal to a public switched telephone network (PSTN) system for outbound communication to the non-VOIP-enabled client or non-client phone. Hence, the telephony server **112** may include a PSTN system and may in some cases access an external PSTN system.

[0031] The telephony server **112** includes one or more session border controllers (SBCs) for interfacing the SIP zone with one or more aspects external to the telephony server **112**. In particular, an SBC can act as an intermediary to transmit and receive SIP requests and responses between clients or non-client devices of a given customer with clients or non-client devices external to that customer. When incoming telephony traffic for delivery to a client of a customer, such as one of the clients **104A** through **104D**, originating from outside the telephony server **112** is received, a SBC receives the traffic and forwards it to a call switch for routing to the client.

[0032] In some implementations, the telephony server **112**, via the SIP zone, may enable one or more forms of peering to a carrier or customer premise. For example, Internet peering to a customer premise may be enabled to ease the migration of the customer from a legacy provider to a service provider operating the telephony server **112**. In another example, private peering to a customer premise may be enabled to leverage a private connection terminating at one end at the telephony server **112** and at the other end at a computing aspect of the customer environment. In yet another example, carrier peering may be enabled to leverage a connection of a peered carrier to the telephony server **112**.

[0033] In some such implementations, a SBC or telephony gateway within the customer environment may operate as an intermediary between the SBC of the telephony server **112** and a PSTN for a peered carrier. When an external SBC is first registered with the telephony server **112**, a call from a client can be routed through the SBC to a load balancer of the SIP zone, which directs the traffic to a call switch of the telephony server **112**. Thereafter, the SBC may be configured to communicate directly with the call switch.

[0034] The web zone receives telephony traffic from a client of a customer, via the SIP zone, and directs same to the application server **108** via one or more Domain Name System (DNS) resolutions. For example, a first DNS within the web zone may process a request received via the SIP zone and then deliver the processed request to a web service which connects to a second DNS at or otherwise associated with the application server **108**. Once the second DNS resolves the request, it is delivered to the destination service at the application server **108**. The web zone may also include a database for authenticating access to a software application for telephony traffic processed within the SIP zone, for example, a softphone.

[0035] The clients **104A** through **104D** communicate with the servers **108** through **112** of the datacenter **106** via the network **114**. The network **114** can be or include, for example, the Internet, a local area network (LAN), a wide area network (WAN), a virtual private network (VPN), or another public or private means of electronic computer communication capable of transferring data between a client and one or more servers. In some implementations, a client can connect to the network **114** via a communal connection point, link, or path, or using a distinct connection point, link, or path. For example, a connection point, link, or path can be wired, wireless, use other communications technologies, or a combination thereof.

[0036] The network **114**, the datacenter **106**, or another element, or combination of elements, of the system **100** can include network hardware such as routers, switches, other network devices, or combinations thereof. For example, the datacenter **106** can include a load balancer **116** for routing

traffic from the network **114** to various servers associated with the datacenter **106**. The load balancer **116** can route, or direct, computing communications traffic, such as signals or messages, to respective elements of the datacenter **106**. For example, the load balancer **116** can operate as a proxy, or reverse proxy, for a service, such as a service provided to one or more remote clients, such as one or more of the clients **104A** through **104D**, by the application server **108**, the telephony server **112**, and/or another server. Routing functions of the load balancer **116** can be configured directly or via a DNS. The load balancer **116** can coordinate requests from remote clients and can simplify client access by masking the internal configuration of the datacenter **106** from the remote clients.

[0037] In some implementations, the load balancer **116** can operate as a firewall, allowing or preventing communications based on configuration settings. Although the load balancer **116** is depicted in FIG. **1** as being within the datacenter **106**, in some implementations, the load balancer **116** can instead be located outside of the datacenter **106**, for example, when providing global routing for multiple datacenters. In some implementations, load balancers can be included both within and outside of the datacenter **106**. In some implementations, the load balancer **116** can be omitted.

[0038] FIG. **2** is a block diagram of an example internal configuration of a computing device **200** of an electronic computing and communications system. In one configuration, the computing device **200** may implement one or more of the client **104**, the application server **108**, the database server **110**, or the telephony server **112** of the system **100** shown in FIG. **1**.

[0039] The computing device **200** includes components or units, such as a processor **202**, a memory **204**, a bus **206**, a power source **208**, peripherals **210**, a user interface **212**, a network interface **214**, other suitable components, or a combination thereof. One or more of the memory **204**, the power source **208**, the peripherals **210**, the user interface **212**, or the network interface **214** can communicate with the processor **202** via the bus **206**.

[0040] The processor **202** is a central processing unit, such as a microprocessor, and can include single or multiple processors having single or multiple processing cores. Alternatively, the processor **202** can include another type of device, or multiple devices, configured for manipulating or processing information. For example, the processor **202** can include multiple processors interconnected in one or more manners, including hardwired or networked. The operations of the processor **202** can be distributed across multiple devices or units that can be coupled directly or across a local area or other suitable type of network. The processor **202** can include a cache, or cache memory, for local storage of operating data or instructions.

[0041] The memory **204** includes one or more memory components, which may each be volatile memory or non-volatile memory. For example, the volatile memory can be RAM (e.g., a DRAM module, such as DDR DRAM). In another example, the non-volatile memory of the memory **204** can be a disk drive, a solid state drive, flash memory, or phase-change memory. In some implementations, the memory **204** can be distributed across multiple devices. For example, the memory **204** can include network-based memory or memory in multiple clients or servers performing the operations of those multiple devices.

[0042] The memory **204** can include data for immediate access by the processor **202**. For example, the memory **204** can include executable instructions **216**, application data **218**, and an operating system **220**. The executable instructions **216** can include one or more application programs, which can be loaded or copied, in whole or in part, from non-volatile memory to volatile memory to be executed by the processor **202**. For example, the executable instructions **216** can include instructions for performing some or all of the techniques of this disclosure. The application data **218** can include user data, database data (e.g., database catalogs or dictionaries), or the like. In some implementations, the application data **218** can include functional programs, such as a web browser, a web server, a database server, another program, or a combination thereof. The operating system **220** can be, for example, Microsoft Windows®, Mac OS X®, or Linux®; an operating

system for a mobile device, such as a smartphone or tablet device; or an operating system for a non-mobile device, such as a mainframe computer.

[0043] The power source **208** provides power to the computing device **200**. For example, the power source **208** can be an interface to an external power distribution system. In another example, the power source **208** can be a battery, such as where the computing device **200** is a mobile device or is otherwise configured to operate independently of an external power distribution system. In some implementations, the computing device **200** may include or otherwise use multiple power sources. In some such implementations, the power source **208** can be a backup battery.

[0044] The peripherals **210** includes one or more sensors, detectors, or other devices configured for monitoring the computing device **200** or the environment around the computing device **200**. For example, the peripherals **210** can include a geolocation component, such as a global positioning system location unit. In another example, the peripherals can include a temperature sensor for measuring temperatures of components of the computing device **200**, such as the processor **202**. In some implementations, the computing device **200** can omit the peripherals **210**.

[0045] The user interface **212** includes one or more input interfaces and/or output interfaces. An input interface may, for example, be a positional input device, such as a mouse, touchpad, touchscreen, or the like; a keyboard; or another suitable human or machine interface device. An output interface may, for example, be a display, such as a liquid crystal display, a cathode-ray tube, a light emitting diode display, virtual reality display, or other suitable display.

[0046] The network interface **214** provides a connection or link to a network (e.g., the network **114** shown in FIG. 1). The network interface **214** can be a wired network interface or a wireless network interface. The computing device **200** can communicate with other devices via the network interface **214** using one or more network protocols, such as using Ethernet, transmission control protocol (TCP), internet protocol (IP), power line communication, an IEEE 802.X protocol (e.g., Wi-Fi, Bluetooth, or ZigBee), infrared, visible light, general packet radio service (GPRS), global system for mobile communications (GSM), code-division multiple access (CDMA), Z-Wave, another protocol, or a combination thereof.

[0047] FIG. 3 is a block diagram of an example of a software platform **300** implemented by an electronic computing and communications system, for example, the system **100** shown in FIG. 1. The software platform **300** is a UCaaS platform accessible by clients of a customer of a UCaaS platform provider, for example, the clients **104A** through **104B** of the customer **102A** or the clients **104C** through **104D** of the customer **102B** shown in FIG. 1. The software platform **300** may be a multi-tenant platform instantiated using one or more servers at one or more datacenters including, for example, the application server **108**, the database server **110**, and the telephony server **112** of the datacenter **106** shown in FIG. 1.

[0048] The software platform **300** includes software services accessible using one or more clients. For example, a customer **302** as shown includes four clients—a desk phone **304**, a computer **306**, a mobile device **308**, and a shared device **310**. The desk phone **304** is a desktop unit configured to at least send and receive calls and includes an input device for receiving a telephone number or extension to dial to and an output device for outputting audio and/or video for a call in progress. The computer **306** is a desktop, laptop, or tablet computer including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The mobile device **308** is a smartphone, wearable device, or other mobile computing aspect including an input device for receiving some form of user input and an output device for outputting information in an audio and/or visual format. The desk phone **304**, the computer **306**, and the mobile device **308** may generally be considered personal devices configured for use by a single user. The shared device **310** is a desk phone, a computer, a mobile device, or a different device which may instead be configured for use by multiple specified or unspecified users.

[0049] Each of the client devices **304** through **310** includes or runs on a computing device configured to access at least a portion of the software platform **300**. In some implementations, the



customer **302** may include additional clients not shown. For example, the customer **302** may include multiple clients of one or more client types (e.g., multiple desk phones or multiple computers) and/or one or more clients of a client type not shown in FIG. 3 (e.g., wearable devices or televisions other than as shared devices). For example, the customer **302** may have tens or hundreds of desk phones, computers, mobile devices, and/or shared devices.

[0050] The software services of the software platform **300** generally relate to communications tools but are in no way limited in scope. As shown, the software services of the software platform **300** include telephony software **312**, conferencing software **314**, messaging software **316**, and other software **318**. Some or all of the software **312** through **318** uses customer configurations **320** specific to the customer **302**. The customer configurations **320** may, for example, be data stored within a database or other data store at a database server, such as the database server **110** shown in FIG. 1.

[0051] The telephony software **312** enables telephony traffic between ones of the client devices **304** through **310** and other telephony-enabled devices, which may be other ones of the client devices **304** through **310**, other VOIP-enabled clients of the customer **302**, non-VOIP-enabled devices of the customer **302**, VOIP-enabled clients of another customer, non-VOIP-enabled devices of another customer, or other VOIP-enabled clients or non-VOIP-enabled devices. Calls sent or received using the telephony software **312** may, for example, be sent or received using the desk phone **304**, a softphone running on the computer **306**, a mobile application running on the mobile device **308**, or using the shared device **310** that includes telephony features.

[0052] The telephony software **312** further enables phones that do not include a client application to connect to other software services of the software platform **300**. For example, the telephony software **312** may receive and process calls from phones not associated with the customer **302** to route that telephony traffic to one or more of the conferencing software **314**, the messaging software **316**, or the other software **318**.

[0053] The conferencing software **314** enables audio, video, and/or other forms of conferences between multiple participants, such as to facilitate a conference between those participants. In some cases, the participants may all be physically present within a single location, for example, a conference room, in which the conferencing software **314** may facilitate a conference between only those participants and using one or more clients within the conference room. In some cases, one or more participants may be physically present within a single location and one or more other participants may be remote, in which the conferencing software **314** may facilitate a conference between all of those participants using one or more clients within the conference room and one or more remote clients. In some cases, the participants may all be remote, in which the conferencing software **314** may facilitate a conference between the participants using different clients for the participants. The conferencing software **314** can include functionality for hosting, presenting scheduling, joining, or otherwise participating in a conference. The conferencing software **314** may further include functionality for recording some or all of a conference and/or documenting a transcript for the conference.

[0054] The messaging software **316** enables instant messaging, unified messaging, and other types of messaging communications between multiple devices, such as to facilitate a chat or other virtual conversation between users of those devices. The unified messaging functionality of the messaging software **316** may, for example, refer to email messaging which includes a voicemail transcription service delivered in email format.

[0055] The other software **318** enables other functionality of the software platform **300**. Examples of the other software **318** include, but are not limited to, device management software, resource provisioning and deployment software, administrative software, third party integration software, and the like. In one particular example, the other software **318** can include software for using a peripheral device to transition between user devices within a video conference. In some such cases, the conferencing software **314** may include some or all of the other software **318**.

[0056] The software **312** through **318** may be implemented using one or more servers, for example, of a datacenter such as the datacenter **106** shown in FIG. **1**. For example, one or more of the software **312** through **318** may be implemented using an application server, a database server, and/or a telephony server, such as the servers **108** through **112** shown in FIG. **1**. In another example, one or more of the software **312** through **318** may be implemented using servers not shown in FIG. **1**, for example, a meeting server, a web server, or another server. In yet another example, one or more of the software **312** through **318** may be implemented using one or more of the servers **108** through **112** and one or more other servers. The software **312** through **318** may be implemented by different servers or by the same server.

[0057] Features of the software services of the software platform **300** may be integrated with one another to provide a unified experience for users. For example, the messaging software **316** may include a user interface element configured to initiate a call with another user of the customer **302**. In another example, the telephony software **312** may include functionality for elevating a telephone call to a conference. In yet another example, the conferencing software **314** may include functionality for sending and receiving instant messages between participants and/or other users of the customer **302**. In yet another example, the conferencing software **314** may include functionality for file sharing between participants and/or other users of the customer **302**. In some implementations, some, or all, of the software **312** through **318** may be combined into a single software application run on clients of the customer, such as one or more of the client devices **304** through **310**.

[0058] FIG. **4** is a block diagram of an example of a system **400** for using a peripheral device **402** to transition between user devices **404A** and **404D** during a video conference **406**. A first conference participant (“user 1”) may be using the user device **404A** to connect to the video conference **406**. The video conference **406** may include other conference participants, such as a second conference participant (“user 2”) using a user device **404B**, and a third conference participant (“user 3”) using a user device **404C**. Each of the user devices **404A** to **404C** may be a client device such as one of the clients **104A** through **104D** shown in FIG. **1** or **304** through **310** shown in FIG. **3**. A user device such as the user devices **404A** to **404C** may execute software (e.g., client-side conferencing software, which could, for example, be via a client application or a web application used to connect to a conference implemented using server-side conferencing software, such as the conferencing software **314** shown in FIG. **3**) and may connect to a server device **408**. The server device **408** may execute software (e.g., server-side conferencing software, such as the conferencing software **314**) to support the video conference **406** between users using the user devices **404A** to **404C**. For example, the server device **408** could be a server at the datacenter **106** shown in FIG. **1**, such as the application server **108** or the telephony server **112**. Although four user devices **404A** to **404D** are shown and described by example, other numbers of user devices may be used with the system **400**.

[0059] The user devices **404A** to **404C** may be used to communicate media content during the video conference **406**. For example, the conference participants may use the user devices to send video data depicting themselves and audio data representing their speech to other conference participants and to receive video and audio data from the other conference participants. The user devices **404A** to **404C** may also connect to peripheral devices for relaying the media content to a conference participant. In this example, the user device **404A** may connect to the peripheral device **402** for relaying the media content to the first conference participant. The peripheral device **402** could be a device configured to present video and/or audio data to the first conference participant, such as a wearable device (e.g., headphones, headset, smartwatch, or glasses) or another device (e.g., display or speakers). The user device **404A** may connect to the peripheral device **402**, for example, via a Bluetooth connection.

[0060] During the video conference **406**, the first conference participant may wish to transition from the user device **404A** to the user device **404D**. For example, the user device **404A** could be a

smartphone that the first conference participant is using while traveling to a location (e.g., while walking or a passenger in a vehicle), and the user device **404D** could be a desktop, laptop, or tablet at the location. In another example, the user device **404A** could be a device associated with a particular space, such as a computer in an office or living room, and the user device **404D** could be another device associated with another space, such as another computer in a conference room or kitchen. To reduce a loss of information associated with the video conference **406**, such as a loss of media content and/or chat messages, the first conference participant may initiate a transition that designates the peripheral device **402** to buffer a portion of media content of the video conference **406**. In some implementations, the first conference participant may initiate the transition and/or designate the peripheral device **402** through a GUI of the user device **404A**.

[0061] Initiating the transition may cause the user device **404A** to transmit an indication of the transition to the peripheral device **402**. The indication may be a message to the peripheral device **402** that indicates the planned transition from one user device to another. The indication can be sent, for example, via the Bluetooth connection between the user device **404A** and the peripheral device **402**. The indication may cause the peripheral device **402** to temporarily buffer a portion of media content of the video conference **406**. For example, the portion of the media content may include video and/or audio data that is transmitted from the user device **404A** to the video conference **406**. The portion of the media content may also include video and/or audio data that is received by the user device **404A** from the video conference **406**. The user device **404A** may then perform, within the video conference **406**, a transition of the first conference participant from the user device **404A** connected to the video conference **406** to the user device **404D** connected to the video conference **406**. Performing the transition may include disconnecting the user device **404A** from the video conference **406** (e.g., this begins the actual transition), connecting the user device **404D** to the video conference **406**, and terminating the communication between the user device **404A** and the user device **404D**. The peripheral device **402** can use the user device **404A** to relay media content to the conference participant before the transition (e.g., via the Bluetooth connection), then use the buffer of the peripheral device **402** to relay the portion of the media content to the conference participant during the transition, then use the user device **404D** to relay media content to the conference participant after the transition (e.g., via a second Bluetooth connection between the user device **404D** and the peripheral device **402**). While the user device **404A** connects to the video conference **406** at a first time, and the user device **404D** connects to the video conference **406** at a second time, the peripheral device **402** does not have to connect to the video conference **406** at any time.

[0062] During the transition, the peripheral device **402** can relay the portion of the media content from the buffer to the conference participant. The peripheral device **402** may adaptively relay the portion of the media content to the conference participant, such as by reducing video quality and/or audio quality of the portion of the media content in a controlled manner. The peripheral device **402** may perform a time alignment between the portion of the media content in the buffer and the media content received by the user device **404D**. After the time alignment, the peripheral device **402** can resume a real-time relay of media content to the conference participant by using the user device **404D** (e.g., this ends the actual transition). This may include the user device **404D** communicating the portion of the media content during the video conference **406**. As a result, the peripheral device **402** can be utilized during the transition between user devices **404A** and **404D** to reduce a loss of information.

[0063] While the first conference participant can use the peripheral device **402**, and the portion of the media content buffered in the peripheral device **402**, to transition from the user device **404A** to the user device **404D**, other conference participants can use other peripheral devices to transition between other user devices in the same way. For example, the second conference participant could use another peripheral device to transition from the user device **404B** to another user device, and the third conference participant could use yet another peripheral device to transition from the user

device **404C** to yet another user device.

[0064] FIG. 5 is a block diagram of an example of transitioning between user devices during a video conference. A system **500** may include a peripheral device **502**, a first user device **504**, and a second user device **506**. The peripheral device **502** could be the peripheral device **402**, the first user device **504** could be the user device **404A**, and the second user device **506** could be the user device **404D** of FIG. 4. A conference participant (e.g., the first conference participant of FIG. 4) may use the first user device **504** to connect to a video conference via a network (e.g., an Ethernet or Wi-Fi network). The conference participant may use the first user device **504** to communicate media content **508** in a media stream **510** during the video conference. The media stream **510** may include video and audio data that is transmitted from the first user device **504** to the video conference and video and audio data that is received by the first user device **504** from the video conference. The conference participant may also use the peripheral device **502** to relay the media content **508**. For example, the peripheral device **502** may exchange a media stream **512** with the first user device **504**. The peripheral device **502** could be a device configured to present video and/or audio data to the conference participant (e.g., a wearable device, such as headphones, a headset, a smartwatch, or glasses, or another device, such as a display or speakers). The first user device **504** may connect to the peripheral device **502**, for example, via Bluetooth.

[0065] With additional reference to FIG. 6, a block diagram illustrates an example of timing **600** associated with transitioning from the first user device **504** to the second user device **506** within the video conference. The first user device **504** may receive, from the conference participant, an input that identifies the second user device **506** and/or the peripheral device **502**. The input may cause the first user device **504** to transmit an indication **514** to the peripheral device **502**. The indication **514** may cause the peripheral device **502** to temporarily buffer a portion **602** of the media content **508** in a memory **516** of the peripheral device **502**. For example, the memory **516** could be RAM implemented by the peripheral device **502**. The portion **602** of the media content **508** may include video and audio data that is transmitted in the media stream **510**. Buffering the portion **602** of the media content **508** may include storing less than a predefined amount of time of the media content **508** received by the first user device **504** in the memory **516**. For example, the memory **516** could be a micro-buffer configured to store less than ten seconds of the media content **508**. The input may also cause the first user device **504** to communicate a handoff request **518** to the second user device **506** to begin the transition from the first user device **504** to the second user device **506**.

[0066] The second user device **506**, based on the handoff request **518**, may transmit a message **520** to the peripheral device **502** to establish communication with the peripheral device **502**. In some cases, the handoff request **518** can establish a new connection between the second user device **506** and the peripheral device **502**. In some cases, the handoff request **518** can establish that an existing connection between the second user device **506** and the peripheral device **502** remains available. The second user device **506** may connect to the peripheral device **502**, for example, via Bluetooth. The peripheral device **502** may acknowledge the message **520** with a return message **522** to the second user device **506**. Based on a successful communication between the second user device **506** and the peripheral device **502**, determined by the message **520** and the return message **522**, the second user device **506** may communicate an acknowledgement **524** of the handoff request **518** to the first user device **504**. Based on a successful buffering of the portion **602** of the media content **508** in the memory **516**, the peripheral device **502** may also communicate an acknowledgement **526** to the first user device **504**. Referring also to FIG. 6, based on the acknowledgement **524** and the acknowledgement **526**, the first user device **504** may disconnect **604** from the video conference while the peripheral device **502** relays the portion **602** of the media content **508** to the conference participant. Disconnecting from the video conference may include the first user device **504** terminating the media stream **510**. In some cases, the first user device **504** may also disconnect from the network (e.g., the Ethernet or Wi-Fi network). The first user device **504** may then communicate a handoff message **528** to the second user device **506** to complete the transition.

[0067] Based on the handoff message **528**, and while the peripheral device **502** relays the portion **602** of the media content **508**, the second user device **506** may connect **606** to the video conference. Connecting to the video conference may include the second user device **506** connecting to the network (e.g., the Ethernet or Wi-Fi network), including without the peripheral device **502** connecting to the same network. Connecting to the video conference may also include the second user device **506** communicating the media content **508** in a media stream **530** during the video conference. The media stream **530** may include video and audio data that is transmitted from the first user device **504** to the video conference and video and audio data that is received by the first user device **504** from the video conference. The second user device **506** may also relay the media content **508** to the peripheral device **502** in a media stream **532** to enable the peripheral device **502** to perform a time alignment between the portion **602** of the media content **508** and the media stream **532**. In at least some cases, the first user device **504** may disconnect **604** from the network and the video conference, and the second user device **506** may connect **606** to the network and the video conference, without the peripheral device **502** connecting to the network.

[0068] Referring also to FIG. **6**, the peripheral device **502** may perform a time alignment **608** between the portion **602** of the media content **508** and the media content **508** that is received by the second user device **506** in the media stream **530**. Based on the time alignment **608**, the peripheral device **502** may adaptively relay the portion **602** of the media content **508** to the conference participant, such as by reducing video quality (e.g., delivering fewer frames per second) and/or audio quality (e.g., delivering fewer kilobits per second) of the portion of the media content **508** in a controlled manner (e.g., a dynamic buffer). Based on the time alignment **608**, the peripheral device **502** may transition **610** from relaying the portion **602** of the media content **508** to relaying the media stream **530**. For example, buffering the portion **602** of the media content **508** could be performed by a transmission of the media content **508** in real time from the first user device **504** to the peripheral device **502**. The peripheral device **502** can fill the memory **516** using real time data while relaying non-real time data to the conference participant during the transition. For example, before the transition, the portion **602** could compress in the memory **516**. Then, during the transition, the portion **602** could be relayed to the conference participant at a normal speed for the video conference. Then, after the transition, the portion **602** could be relayed at a speed based on the time alignment **608**. This could result in a temporary reduction in playback speed (e.g., 5%) that would likely be unnoticeable to the conference participant while still accommodating the transition. For example, this could include speeding up the media content (e.g., going from 10 frames per second to 12 frames per second) or slowing down the media content (e.g., dropping every other frame) before returning to real time delivery of media content to the conference participant. The peripheral device **502** may also delete the portion **602** of the media content **508** from the memory **516** after the transition **610**, such as by overwriting the memory **516**. The second user device **506** may also terminate communications with the first user device **504** after the transition **610**.

[0069] In some implementations, the portion **602** of the media content **508** may include metadata associated with the video conference, such as chat messages and/or emoji based reactions associated with the video conference. This may enable maintaining the metadata (e.g., the chat messages and/or emoji based reactions) during the transition from the first user device **504** to the second user device **506**. As a result, the conference participant can view information associated with the video conference (e.g., the chat messages and/or emoji based reactions, exchanged between conference participants before the transition **610**, after the transition **610**) as though the conference participant never left the video conference.

[0070] In some implementations, an authorization of the first user device **504** may enable the first user device **504** to connect to the video conference. For example, the authorization could be a permission or credential from a server device (e.g., the server device **408**). The authorization could be required to enable the first user device **504** to join the video conference. The conference participant can extend the authorization to the second user device **506** based on the authorization of

the first user device **504** and the connection between the first user device **504** and the peripheral device **502**. For example, the conference participant can use the peripheral device **502** to transfer the authorization from the first user device **504** to the second user device **506**. As a result, the second user device **506** can join the video conference without obtaining the authorization from the server device. In some implementations, the peripheral device **502** could be a wearable technology, such as augmented reality (AR) glasses. The conference participant can look at the second user device **506**, while using the AR glasses and the first user device **504** to connect to the video conference, to selectively authorize the second user device **506** to connect to the video conference. [0071] In some implementations, a machine learning model may be used to determine the portion **602** of the media content **508** to buffer in the memory **516** (e.g., the predefined amount of time), the second user device **506** for transitioning, and/or the peripheral device **502**. The machine learning may be trained using historical information, including past selections of user devices, past selections of peripheral devices, and/or past buffering during transitions. For example, the machine learning model can be trained using a training data set including data samples representing parameters for historical transitions. The training data set can enable the machine learning model to learn patterns, such as particular user devices, particular peripheral devices, and particular buffering. The training can be periodic, such as by updating the machine learning model on a discrete time interval basis (e.g., once per week or month), or otherwise. The training data set may derive from multiple conference participants or may be specific to a particular conference participant (e.g., the first conference participant). The machine learning model may, for example, be or include one or more of a neural network (e.g., a convolutional neural network, recurrent neural network, deep neural network, or other neural network), decision tree, vector machine, Bayesian network, cluster-based system, genetic algorithm, deep learning system separate from a neural network, or other machine learning model.

[0072] FIG. 7 is an illustration of an example of a GUI **700** output for display during a video conference. The GUI **700** could be configured for display to a user interface of a user device, such as the user device **404A** of FIG. 4, or the first user device **504** of FIG. 5. For example, the GUI **700** could be configured for display to a user interface like the user interface **212**. The GUI **700** may include user tiles associated with conference participants, such as a user tile **702A** associated with the first conference participant (“user 1”) using user device **404A**, a user tile **702B** associated with the second conference participant (“user 2”) using user device **404C**, and a user tile **702C** associated with the third conference participant (“user 3”) using user device **404D** of FIG. 4. The GUI **700** may also include a history of chat messages **704** that may be typed and exchanged by the conference participants during the video conference. The GUI **700** may also include emoji based reactions **706** exchanged between the conference participants.

[0073] The GUI **700** could be output to the user device **404A** (or the first user device **504**) of the first conference participant at a first time. The first conference participant may have a peripheral device **708**, such as a headset visible in the user tile **702A**. The first conference participant may transmit (e.g., from the user device **404A**) an indication to the peripheral device **708** to temporarily buffer a portion of media content of the video conference. When buffering is complete, the GUI **700** may display a message **710** indicating buffering is complete and requesting input from the first conference participant to complete the transition (“Yes” or “No”). When the first conference participant is ready to transition (“Yes”), the peripheral device **708** may then be used to perform, within the video conference, the transition from the user device **404A** to the user device **404D** (or the second user device **506**). Performing the transition may include causing the peripheral device **708** to relay the portion of the media content during the transition. The portion of the media content may include video and audio data, including the video data represented by the user tiles **702A** to **702C**, and the audio data associated with the user tiles **702A** to **702C**. The portion of the media content may also include metadata associated with the video conference, such as the history of chat messages **704** and the emoji based reactions **706**. The GUI **700** could then be output to the user

device **404D** at a second time, including with the history of chat messages **704** and the emoji based reactions **706** preserved the buffering.

[0074] To further describe some implementations in greater detail, reference is next made to examples of techniques which may be performed using a peripheral device. FIG. **8** is a flowchart of an example of a technique **800** for buffering a portion of media content in a peripheral device. The technique **800** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-7**. The technique **800** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **800** or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

[0075] For simplicity of explanation, the technique **800** is depicted and described herein as a series of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

[0076] At **802**, a system may receive, during a video conference, a first input to transition from a first user device connected to a video conference to a second user device. For example, the system **500** of FIG. **5**, via a GUI of the first user device **504**, may receive the first input. The first input may be provided by a conference participant (e.g., the first conference participant of FIG. **4**, or “user 1”) that wishes to transition from the first user device to the second user device (e.g., the second user device **506**). In some implementations, a machine learning model can predict the first input. The machine learning model could run on the first user device.

[0077] At **804**, the system may receive, during the video conference, a second input designating a peripheral device. For example, the system **500** of FIG. **5**, via a GUI of the first user device **504**, may receive the second input. The second input may be provided by a conference participant that wishes to transition from the first user device to the second user device by utilizing the peripheral device (e.g., the peripheral device **402**, or the peripheral device **502**). The peripheral device may be a device that the conference participant is using during the video conference, such as a wearable device (e.g., headphones, headset, smartwatch, or glasses) or another device configured to present video and/or audio data to the conference participant (e.g., display or speakers). The first user device may be connected to the peripheral device via a short-range wireless connection such as Bluetooth. In some implementations, a machine learning model can predict the second input. The machine learning model could run on the first user device.

[0078] At **806**, the system may transmit, from the first user device connected to the video conference and associated with the conference participant, an indication to the peripheral device to temporarily buffer a portion of media content of the video conference. For example, the first user device **504** could transmit the indication **514** to the peripheral device **502** to temporarily buffer the portion **602** of the media content **508** of the video conference. The indication may cause the peripheral device to temporarily buffer the portion of the media content of the video conference. The portion of the media content may include video and audio data that is transmitted via the first user device. The portion of the media content may also include metadata associated with the video conference, such as chat messages and emoji based reactions. Buffering the portion of the media content may include storing less than a predefined amount of time of the media content in a memory of the peripheral device (e.g., a RAM, such as a static or dynamic RAM). In some implementations, a machine learning model can predict the portion of the media content to buffer in the memory. The machine learning model could run on the first user device and/or the peripheral device.

[0079] At **808**, the system may establish a connection between the second user device and the

peripheral device. The connection may be established based on the first input, the second input, and/or the indication. The connection could be another short-range wireless connection such as Bluetooth. For example, the connection could be established via the message **520** and the return message **522** of FIG. **5**. Establishing the connection may then enable a transition between user devices (e.g., from the first user device to the second user device) to be performed within the video conference.

[0080] FIG. **9** is a flowchart of an example of a technique **900** for using a peripheral device to transition between user devices within a video conference. The technique **900** can be executed using computing devices, such as the systems, hardware, and software described with respect to FIGS. **1-7**. The technique **900** can be performed, for example, by executing a machine-readable program or other computer-executable instructions, such as routines, instructions, programs, or other code. The steps, or operations, of the technique **900** or another technique, method, process, or algorithm described in connection with the implementations disclosed herein can be implemented directly in hardware, firmware, software executed by hardware, circuitry, or a combination thereof.

[0081] For simplicity of explanation, the technique **900** is depicted and described herein as a series of steps or operations. However, the steps or operations in accordance with this disclosure can occur in various orders and/or concurrently. Additionally, other steps or operations not presented and described herein may be used. Furthermore, not all illustrated steps or operations may be required to implement a technique in accordance with the disclosed subject matter.

[0082] At **902**, a system may buffer a portion of media content of a video conference in a peripheral device. For example, the system **500** of FIG. **5**, may buffer the portion **602** of the media content **508** of the video conference in the peripheral device **502**. The system may buffer the portion of the media content based on the technique **800** described in FIG. **8**, including step **806**. For example, the system may buffer the portion of the media content in a memory of the peripheral device based on an indication from the first user device (e.g., the first user device **504**). In some implementations, a machine learning model can predict the portion of the media content to buffer in the memory. The machine learning model could run on the first user device and/or the peripheral device. While the system buffers the portion, a second user device (e.g., the second user device **506**) may establish a connection with the peripheral device.

[0083] At **904**, the system may determine if a conference participant (e.g., the first conference participant of FIG. **4**, or “user 1”) is ready to transition from the first user device to the second user device. For example, the system, via a GUI of the first user device, may display to the conference participant that the system is ready to perform the transition (e.g., the message **710**, displayed based on completing a buffering of the portion of the media content and/or establishing a connection between the second user device and the peripheral device) and may wait to receive an input. If the system does not receive an input from the conference participant to perform the transition (“No”), the system can wait, returning to **902** to update the portion of the media content stored in the peripheral device with current media content associated with the video conference. If the system does receive an input to perform the transition (“Yes”), the system can continue to **906**.

[0084] At **906**, the system may perform, within the video conference, a transition of the conference participant from the first user device to the second user device connected to the video conference. The transition may include causing the peripheral device to relay the portion of the media content during the transition. For example, the transition may include the peripheral device **502** relaying the portion **602** of the media content **508** during the transition. In some implementations, the peripheral device may perform a time alignment between the portion of the media content and the media content that is received by the second user device in the media stream. Based on the time alignment, the peripheral device may adaptively relay the portion of the media content to the conference participant, such as by reducing video quality and/or audio quality of the portion of the media content in a controlled manner. Based on the time alignment, the peripheral device may transition from relaying the portion of the media content to relaying the media stream in real time.



[0085] At **908**, the system may terminate communications between the first user device and the second user device. The system may also delete the portion of the media content from the memory of the peripheral device. For example, the system may delete the portion of the media content by overwriting the memory in the peripheral device.

[0086] Some implementations may include a method that includes transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of media content of the video conference; and performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference including causing the peripheral device to relay the portion of the media content during the transition. In some implementations, the media content includes data transmitted from the first user device. In some implementations, the peripheral device is a wearable device associated with the conference participant. In some implementations, the method may include buffering the portion of the media content by storing less than a predefined amount of time of the media content received by the first user device in a RAM of the peripheral device. In some implementations, the method may include relaying the portion of the media content to the conference participant during the transition with the portion of the media content being reduced in at least one of video quality or audio quality. In some implementations, the method may include disconnecting the first user device from a network and the video conference; and connecting the second user device to the network and the video conference without connecting the peripheral device to the network. In some implementations, the method may include maintaining chat messages associated with the video conference during the transition from the first user device to the second user device by including the chat messages in the portion of the media content. In some implementations, the method may include receiving an input that identifies the second user device and the peripheral device, the input causing a communication between the second user device and the peripheral device before the transition from the first user device to the second user device. In some implementations, the method may include allowing the second user device to connect to the video conference based on an authorization of the first user device to connect to the video conference and a connection between the first user device and the peripheral device. In some implementations, the method may include determining the portion of the media content to buffer by using a machine learning model.

[0087] Some implementations may include an apparatus that includes a memory and a processor configured to execute instructions stored in the memory to transmit, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of media content of the video conference; and perform, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference including causing the peripheral device to relay the portion of the media content during the transition. In some implementations, the processor is further configured to execute instructions stored in the memory to determine the second user device using a machine learning model. In some implementations, the processor is further configured to execute instructions stored in the memory to buffer the portion of the media content by storing less than a ten seconds of the media content received by the first user device in a micro-buffer of the peripheral device. In some implementations, the processor is further configured to execute instructions stored in the memory to delete the portion of the media content after the transition. In some implementations, the peripheral device is a headset.

[0088] Some implementations may include a non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of media content of the video conference; and performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference

including causing the peripheral device to relay the portion of the media content during the transition. In some implementations, the operations further comprise communicating a handoff event between the first user device and the second user device. In some implementations, the operations further comprise terminating a communication between the first user device and the second user device after the transition. In some implementations, the operations further comprise performing a time alignment between the portion of the media content and the media content received by the second user device. In some implementations, the peripheral device comprises AR glasses.

[0089] The implementations of this disclosure can be described in terms of functional block components and various processing operations. Such functional block components can be realized by a number of hardware or software components that perform the specified functions. For example, the disclosed implementations can employ various integrated circuit components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which can carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, where the elements of the disclosed implementations are implemented using software programming or software elements, the systems and techniques can be implemented with a programming or scripting language, such as C, C++, Java, JavaScript, assembler, or the like, with the various algorithms being implemented with a combination of data structures, objects, processes, routines, or other programming elements.

[0090] Functional aspects can be implemented in algorithms that execute on one or more processors. Furthermore, the implementations of the systems and techniques disclosed herein could employ a number of conventional techniques for electronics configuration, signal processing or control, data processing, and the like. The words “mechanism” and “component” are used broadly and are not limited to mechanical or physical implementations, but can include software routines in conjunction with processors, etc. Likewise, the terms “system” or “tool” as used herein and in the figures, but in any event based on their context, may be understood as corresponding to a functional unit implemented using software, hardware (e.g., an integrated circuit, such as an ASIC), or a combination of software and hardware. In certain contexts, such systems or mechanisms may be understood to be a processor-implemented software system or processor-implemented software mechanism that is part of or callable by an executable program, which may itself be wholly or partly composed of such linked systems or mechanisms.

[0091] Implementations or portions of implementations of the above disclosure can take the form of a computer program product accessible from, for example, a computer-usable or computer-readable medium. A computer-usable or computer-readable medium can be a device that can, for example, tangibly contain, store, communicate, or transport a program or data structure for use by or in connection with a processor. The medium can be, for example, an electronic, magnetic, optical, electromagnetic, or semiconductor device.

[0092] Other suitable mediums are also available. Such computer-usable or computer-readable media can be referred to as non-transitory memory or media and can include volatile memory or non-volatile memory that can change over time. The quality of memory or media being non-transitory refers to such memory or media storing data for some period of time or otherwise based on device power or a device power cycle. A memory of an apparatus described herein, unless otherwise specified, does not have to be physically contained by the apparatus, but is one that can be accessed remotely by the apparatus, and does not have to be contiguous with other memory that might be physically contained by the apparatus.

[0093] The present teachings provide: a method including transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference. Performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference. Disconnecting the first user device from the video

conference before the second user device is connected to the video conference. Transitioning the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

[0094] The present teachings provide: an apparatus with memory and a processor. The processor is configured to execute instructions stored in the memory. The processor is configured to transmit, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference. The processor is configured to perform, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference. The processor is configured to disconnect the first user device from the video conference before the second user device is connected to the video conference. The processor is configured to transition the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

[0095] The present teachings provide: a non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations. The operations include transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference. The operations include performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference. The operations include: disconnecting the first user device from the video conference before the second user device is connected to the video conference. The operations include: transitioning the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

[0096] While the disclosure has been described in connection with certain implementations, it is to be understood that the disclosure is not to be limited to the disclosed implementations but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

## Claims

1. A method, comprising: transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference; performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference; disconnecting the first user device from the video conference before the second user device is connected to the video conference; and transitioning the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.
2. The method of claim 1, wherein the portion of the video conference includes data transmitted from the first user device.
3. The method of claim 1, wherein the peripheral device is a wearable device associated with the conference participant.
4. The method of claim 1, further comprising: buffering the portion of the video conference by storing the portion of the video conference received by the first user device in a random access memory (RAM) of the peripheral device.
5. The method of claim 1, further comprising: indicating to the peripheral device to temporarily buffer the portion of the video conference in preparation for the first user device disconnecting from the video conference.
6. The method of claim 1, further comprising: compressing the portion of the video conference

before transitioning to the second user device.

**7.** The method of claim 1, further comprising: maintaining chat messages associated with the video conference during the transition from the first user device to the second user device by including the chat messages in the portion of the video conference transitioned to the peripheral device.

**8.** The method of claim 1, further comprising: receiving an input that identifies the second user device and the peripheral device, the input causing a communication between the second user device and the peripheral device before the transition from the first user device to the second user device.

**9.** The method of claim 1, further comprising: allowing the second user device to connect to the video conference based on an authorization of the first user device to connect to the video conference and a connection between the first user device and the peripheral device.

**10.** The method of claim 1, further comprising: performing a time alignment from the peripheral device to transition the portion of the video conference to a real-time media stream of the video conference.

**11.** An apparatus, comprising: a memory; and a processor configured to execute instructions stored in the memory to: transmit, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference; perform, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference; disconnect the first user device from the video conference before the second user device is connected to the video conference; and transition the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

**12.** The apparatus of claim 11, wherein the processor is further configured to execute instructions stored in the memory to: communicate a handoff request from the first user device to the second user device to begin the transition of the conference participant from the first user device to the second user device.

**13.** The apparatus of claim 11, wherein the processor is further configured to execute instructions stored in the memory to: buffering the portion of the video conference by storing the portion of the video conference received by the first user device in a random access memory (RAM) of the peripheral device.

**14.** The apparatus of claim 11, wherein the processor is further configured to execute instructions stored in the memory to: delete the portion of the video conference after the transition to the second user device.

**15.** The apparatus of claim 11, wherein a handoff request establishes a new connection between the second user device and the peripheral device.

**16.** A non-transitory computer readable medium storing instructions operable to cause one or more processors to perform operations comprising: transmitting, from a first user device connected to a video conference and associated with a conference participant, an indication to a peripheral device to buffer a portion of the video conference; performing, within the video conference, a transition of the conference participant from the first user device to a second user device connected to the video conference; disconnecting the first user device from the video conference before the second user device is connected to the video conference; and transitioning the video conference to the second user device while displaying the buffer of the portion of the video conference until the second user device is aligned with the video conference.

**17.** The non-transitory computer readable medium storing instructions of claim 16, wherein the operations further comprise: communicating a handoff event between the first user device and the second user device.

**18.** The non-transitory computer readable medium storing instructions of claim 16, wherein the operations further comprise: terminating a communication between the first user device and the

second user device after the transition.

**19.** The non-transitory computer readable medium storing instructions of claim 16, wherein the operations further comprise: performing a time alignment between the portion of the video conference and media content received by the second user device.

**20.** The non-transitory computer readable medium storing instructions of claim 16, wherein a handoff request establishes a new connection between the second user device and the peripheral device.

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